```
  A new version of Flutter is available!

  To update to the latest version, run "flutter upgrade".

Doctor summary (to see all details, run flutter doctor -v):
[✓] Flutter (Channel stable, 3.47.3, on Microsoft Windows [Version 10.0.26200.9457], locale en-US)
[✓] Windows Version (Windows 11 or higher, 25H2, 2009)
[✓] Android toolchain - develop for Android devices (Android SDK version 36.0.0)
[✓] Chrome - develop for the web
[✓] Visual Studio - develop Windows apps (Visual Studio Community 2026 18.10.0)
[✓] Connected device (3 available)
[✓] Network resources

• No issues found!

C:\Users\Admin>flutter create my_first_app
Creating project my_first_app...
Resolving dependencies in `my_first_app`... (2.8s)
Downloading packages...
Got dependencies in `my_first_app`.
Wrote 131 files.

All done!
You can find general documentation for Flutter at: https://docs.flutter.dev/
Detailed API documentation is available at: https://api.flutter.dev/
If you prefer video documentation, consider: https://www.youtube.com/c/flutterdev

In order to run your application, type:

  $ cd my_first_app
  $ flutter run

Your application code is in my_first_app\lib\main.dart.


C:\Users\Admin>cd my_first_app

C:\Users\Admin\my_first_app>flutter devices
Found 3 connected devices:
  Windows (desktop) • windows • windows-x64    • Microsoft Windows [Version 10.0.26200.9457]
  Chrome (web)      • chrome  • web-javascript • Google Chrome 154.0.8037.58
  Edge (web)        • edge    • web-javascript • Microsoft Edge 154.0.4258.37

Run "flutter emulators" to list and start any available device emulators.
```

Hot Reload

Hot Reload applies changes made to the Dart code to the running Flutter application without completely restarting the application. It usually preserves the current application state.

It is used when;

You are making small changes to the UI, styling, layout, or Dart code and want to see the result immediately while continuing from the current state of the application.

Hot Restart

Hot Restart completely restarts the Flutter application and rebuilds its state from the beginning. Unlike Hot Reload, the current application state is lost.

It is used;

Changes affect initialization, global/static state, or other code where Hot Reload does not properly apply the changes, or when you want to reset the application to its initial state.

TASK 2

```
Server version: 8.0.46 MySQL Community Server - GPL

Copyright (c) 2000, 2026, Oracle and/or its affiliates.

Oracle is a registered trademark of Oracle Corporation and/or its
affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement

mysql> CREATE DATABASE school;
ERROR 1007 (HY000): Can't create database 'school'; database exists
mysql> DROP DATABASE school;
Query OK, 1 row affected (0.22 sec)

mysql> CREATE DATABASE school;
Query OK, 1 row affected (0.01 sec)

mysql> USE school;
Database changed
mysql> CREATE TABLE students (
    ->     id INT AUTO_INCREMENT PRIMARY KEY,
    ->     name VARCHAR(100),
    ->     email VARCHAR(150) UNIQUE,
    ->     enrolled_on DATE
    -> );
Query OK, 0 rows affected (0.10 sec)

mysql> INSERT INTO students (name, email, enrolled_on)
    -> VALUES
    -> ('John Doe', 'john.doe@example.com', '2026-09-30'),
    -> ('Jane Smith', 'jane.smith@example.com', '2026-09-30');
Query OK, 2 rows affected (0.04 sec)
Records: 2  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM students;
+----+------------+-----------------------+-------------+
| id | name       | email                 | enrolled_on |
+----+------------+-----------------------+-------------+
|  1 | John Doe   | john.doe@example.com  | 2026-09-30  |
|  2 | Jane Smith | jane.smith@example.com | 2026-09-30  |
+----+------------+-----------------------+-------------+
2 rows in set (0.01 sec)

mysql>
```

EXPLANATION

Using the MySQL root account for an application is not a good security practice because root has almost complete control over the database server. If the application is hacked or its password is exposed, an attacker could use those privileges to delete databases, change settings, create other users, or access information that the application should not be able to access.

A safer approach is to create a separate database user specifically for the application. This user should only have the permissions it actually needs, such as viewing, adding, updating, and deleting student records. This follows the principle of least privilege, which helps reduce the damage that could occur if the application's credentials were ever compromised.
Available next action: Create a downloadable DOCX file here in this chat containing the editable prose above

TASK 3

```
(venv) C:\Users\Admin\my_first_app\python_setup_lab\python_setup_lab>venv\Scripts\activate

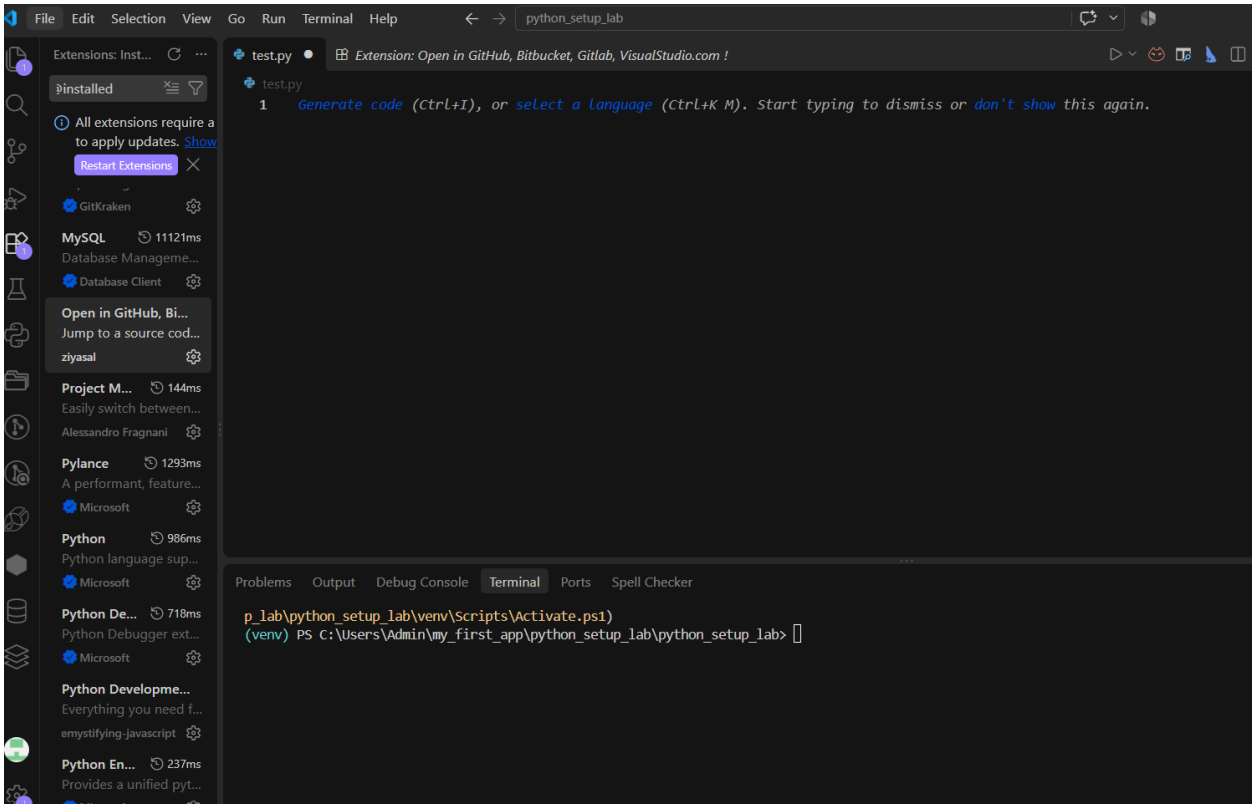
(venv) C:\Users\Admin\my_first_app\python_setup_lab\python_setup_lab>pip install requests
Collecting requests
  Using cached requests-2.34.2-py3-none-any.whl.metadata (4.8 kB)
Collecting charset_normalizer<4,>=2 (from requests)
  Using cached charset_normalizer-3.5.2-cp313-cp313-win_amd64.whl.metadata (47 kB)
Collecting idna<4,>=2.5 (from requests)
  Using cached idna-3.20-py3-none-any.whl.metadata (7.2 kB)
Collecting urllib3<3,>=1.26 (from requests)
  Using cached urllib3-2.8.0-py3-none-any.whl.metadata (7.4 kB)
Collecting certifi>=2023.5.7 (from requests)
  Using cached certifi-2026.7.22-py3-none-any.whl.metadata (2.5 kB)
Using cached requests-2.34.2-py3-none-any.whl (73 kB)
Using cached charset_normalizer-3.5.2-cp313-cp313-win_amd64.whl (206 kB)
Using cached idna-3.20-py3-none-any.whl (69 kB)
Using cached urllib3-2.8.0-py3-none-any.whl (135 kB)
Using cached certifi-2026.7.22-py3-none-any.whl (136 kB)
Installing collected packages: urllib3, idna, charset_normalizer, certifi, requests
Successfully installed certifi-2026.7.22 charset_normalizer-3.5.2 idna-3.20 requests-2.34.2 urllib3-2.8.0

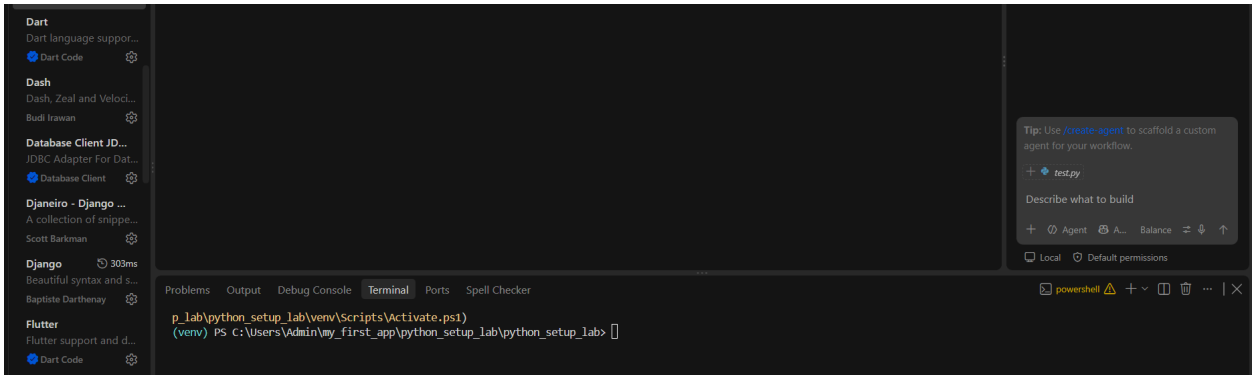
(venv) C:\Users\Admin\my_first_app\python_setup_lab\python_setup_lab>pip list
Package            Version
------------------ ---------
certifi            2026.7.22
charset-normalizer 3.5.2
idna               3.20
pip                26.2.1
requests           2.34.2
urllib3            2.8.0

(venv) C:\Users\Admin\my_first_app\python_setup_lab\python_setup_lab>python -m pip freeze > requirements.txt

(venv) C:\Users\Admin\my_first_app\python_setup_lab\python_setup_lab>type requirements.txts
```

TASK 4


File Edit Selection View Go Run Terminal Help
python_setup_lab
Extensions: Inst...
test.py
Extension: Open in GitHub, Bitbucket, Gitlab, VisualStudio.com !
1 Generate code (Ctrl+I), or select a language (Ctrl+K M). Start typing to dismiss or don't show this again.
All extensions require a to apply updates. Show
Restart Extensions
GitKraken
MySQL 11121ms
Database Manageme...
Database Client
Open in GitHub, Bi...
Jump to a source cod...
ziyasal
Project M... 144ms
Easily switch between...
Alessandro Fragnani
Pylance 1293ms
A performant, feature...
Microsoft
Python 986ms
Python language sup...
Microsoft
Python De... 718ms
Python Debugger ext...
Microsoft
Python Developme...
Everything you need f...
emystifying-javascript
Python En... 237ms
Provides a unified pyt...
Problems Output Debug Console Terminal Ports Spell Checker
p_lab\python_setup_lab\venv\Scripts\Activate.ps1)
(venv) PS C:\Users\Admin\my_first_app\python_setup_lab\python_setup_lab>


Dart
Dart language suppor...
Dart Code
Dash
Dash, Zeal and Veloci...
Budi Irawan
Database Client JD...
JDBC Adapter For Dat...
Database Client
Djaneiro - Django ...
A collection of snippe...
Scott Barkman
Django 303ms
Beautiful syntax and s...
Baptiste Darthenay
Flutter
Flutter support and d...
Dart Code
Tip: Use /create-agent to scaffold a custom agent for your workflow.
test.py
Describe what to build
Agent
Balance
Local
Default permissions
Problems Output Debug Console Terminal Ports Spell Checker
powershell
p_lab\python_setup_lab\venv\Scripts\Activate.ps1)
(venv) PS C:\Users\Admin\my_first_app\python_setup_lab\python_setup_lab>